ParcelCare | Autonomous Recovery Workflow

Mentor discussion brief · Updated proposal: AI recommends first, authorized humans opt in, then the agent executes and follows up.

01 System workflow

1 · Exception reported

Customer complaint OR rider failed-attempt report; tracking ID, details and optional photo evidence

2 · Create or match case

Use tracking ID plus issue context to merge duplicate reports; different issues can remain separate cases

3 · AI investigates and recommends

Review shipment data, evidence, priority, permissions and available recovery actions; show proposed next step to office staff

4 · Staff chooses whether to activate AI recovery

Staff approves the suggested service/action; if declined, case stays on manual handling route

5 · AI executes within approved limits

Contacts customer or seller as applicable, uses validated backend tools, creates tasks and sends updates

6 · Physical work + verification

Rider/operations complete assigned work; AI monitors statuses, escalates blockers and closes only after verified outcome

Key distinction

Office-staff approval activates the proposed AI service; it does not replace customer consent, seller authorization, courier policy checks, or physical delivery confirmation.

ParcelCare | Case-Specific Autonomy

02 What the AI does after staff approval

A · Wrong address — courier-controlled recovery

AI proposes address recovery

Staff sees: “I can contact the customer, confirm a new address, and coordinate redelivery.”

Staff approves AI contacts customer

Customer supplies and confirms corrected address; AI checks delivery zone, parcel state and available slots

AI updates and schedules

Only permitted changes are applied; backend creates a redelivery task and notifies customer/rider

Rider delivers AI verifies outcome

Track task completion and delivery proof; unresolved or invalid requests go to staff

B · Wrong item / damaged parcel — seller-dependent recovery

AI proposes seller coordination

Staff sees: “I can collect evidence and contact the registered seller for return/replacement authorization.”

Staff approves AI contacts verified seller

For demo: teammate replies in an internal Seller Chat Simulator, not through real SMS

AI checks seller decision and required permissions

Seller may approve return/replacement, request evidence, decline or defer; uncertain liability or unclear authority is escalated

AI creates authorized operational tasks

Return pickup / replacement transport only when prerequisites are satisfied; rider handles physical collection and delivery

AI tracks outcome and escalates exceptions

Replacement “promised” is not “ready”; case closes only after required return/delivery confirmations

AI autonomy boundary

AI can investigate, message, interpret replies, plan and invoke controlled tools. It cannot independently approve refunds, decide contested liability, invent seller consent, or mark an unfinished shipment as resolved.

ParcelCare | Architecture & Business Proposal

03 Practical 2-day MVP stack

Frontend	React / Next.js + TypeScript + Tailwind + shadcn/ui; Customer, Rider and Office Staff dashboards; internal Seller Simulator for demo.
Backend	FastAPI (Python): case APIs, authorization gates, shipment operations, event-driven agent triggers and tool execution.
AI	OpenAI API with function calling; persistent case state; AI produces recommendations and invokes only approved tools.
Data + evidence	Supabase PostgreSQL for shipments, cases, messages, approvals, tasks, events; Supabase Storage for evidence; optional Realtime.
Deployment	Next.js on Vercel; FastAPI on a Python-capable host; Supabase hosted project. Use test data and mock dispatch/seller services.

Technical interaction

Next.js dashboards + Seller Simulator

Complaint, rider report, staff approval, seller reply, case timeline

FastAPI + policy/permission checks

Creates or matches cases; records approvals; validates AI tool calls; writes task updates

OpenAI Agent Supabase + backend tools

Investigates, suggests, communicates, resumes on new events; updates only through controlled FastAPI actions

04 Paid business-plan concept (proposal, not finalized)

Base plan	Manual exception management, dashboards, case visibility and status tracking.
AI Recovery plan	AI recommendations, staff-approved recovery execution, seller/customer follow-up and audit trail.
Future advanced plan	Pre-approved policy automation for eligible low-risk cases, with spending/authority limits and exception-only staff review.

Commercial and product considerations

Charge the logistics company, not the customer, for an optional AI recovery capability. Test whether customers value reduced handling time and higher recovery completion enough to pay. Pricing is not yet validated. Requiring staff approval for every case may reduce automation benefits; policy-based opt-in is a potential next step.

48-hour MVP boundary

Build: one working wrong-address flow, seller-chat approval simulation for wrong item, damage evidence/escalation, 3 dashboards, approval gate, task creation and audit log. Exclude: real SMS, live warehouse integrations, refunds, automated liability judgments and production-grade billing.